CSCI 271: Assignment 4 — Iteration & Loop Patterns

Part 1: Multiple Choice (8 Questions · 8 points)

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1. In a while loop, when is the condition checked?

☐ A) After the loop body runs, but not before
☐ B) Before every single pass, including the first
☐ C) Only once, before the loop starts
☐ D) Only after the loop has already ended

2. Which loop pattern do you use when you don't know in advance how many values the user will enter?

☐ A) Counter-controlled
☐ B) Sentinel-controlled
☐ C) do...while only
☐ D) None of these

3. In a sentinel loop, why do we check if (count != 0) before dividing to get an average?

☐ A) To silence a compiler warning
☐ B) To avoid dividing by zero when no values were entered
☐ C) To make the loop run faster
☐ D) It's required syntax for every while loop

4. Which for loop header correctly prints the numbers 1 through 5?

☐ A) for (int i = 0; i < 5; i++)
☐ B) for (int i = 1; i <= 5; i++)
☐ C) for (int i = 1; i < 5; i++)
☐ D) for (int i = 5; i <= 1; i++)

5. What is different about a do...while loop compared to a while loop?

☐ A) do...while never uses a condition
☐ B) do...while's body always runs at least once
☐ C) do...while can only count downward
☐ D) do...while doesn't require a semicolon anywhere

6. What does continue; do inside a loop?

☐ A) Immediately exits the loop entirely
☐ B) Skips the rest of the current pass and moves to the next condition check
☐ C) Restarts the whole program
☐ D) Does the exact same thing as break;

7. Which mistake causes an infinite loop?

☐ A) Using <= instead of < in the condition
☐ B) Forgetting to update the loop's counter or condition variable
☐ C) Declaring the counter as an int
☐ D) Using a for loop instead of a while loop

8. Given: int count = 0; for (int i = 1; i <= 10; i += 2) { count++; } What is count after the loop? [DAY 2] *(5 pt)*

☐ A) 10 ☐ B) 5 ☐ C) 9 ☐ D) 2

Part 2: Coding Problems (22 points)

Problem 1: Multiplication Table (10 pts)

Write a C++ program called mult_table.cpp that prints a multiplication table for a number the user enters.

- ☐ Ask the user to enter a whole number using cin.
- ☐ Use a for loop to multiply that number by 1 through 10.
- ☐ Print one line per multiplication, e.g. “7 x 1 = 7”, all the way through “7 x 10 = 70”.

Example interaction:

```
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
...
7 x 10 = 70
```

Problem 2: Menu Choice Validation (12 pts)

Write a C++ program called menu_validate.cpp that asks the user to enter a menu choice and won't move on until they enter a valid one.

- ☐ Use a do...while loop, since the prompt must show at least once no matter what.
- ☐ Ask the user to enter a choice: 1, 2, or 3.
- ☐ If the number entered is NOT 1, 2, or 3, print “Invalid choice, try again.” and re-prompt.
- ☐ Once a valid choice is entered, print “You selected option X.” with the number they chose.

Example interaction:

```
Enter a menu choice (1, 2, or 3): 5
Invalid choice, try again.
Enter a menu choice (1, 2, or 3): 0
Invalid choice, try again.
Enter a menu choice (1, 2, or 3): 2
You selected option 2.
```